

Arpitha Singh

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in LLM applications, Retrieval-Augmented Generation (RAG), and NLP pipelines, with production experience building AI-powered backend systems using Python, FastAPI, and SQL. Experienced in prompt engineering, intent classification, recommendation systems, and integrating LLMs into scalable REST APIs. Seeking AI Engineer, ML Engineer, or LLM Engineer roles focused on building production-grade AI systems.

EXPERIENCE

Galactix Solutions Pvt Ltd

Remote

AI/ML Developer Intern

Jan 2026 – July 2026

- Engineered a Retrieval-Augmented Generation (RAG) pipeline integrating GPT-4o for the KodeMi Labs EdTech platform, applying prompt engineering to improve response relevance and reduce hallucinations across Guest, Logged-in, and Subscribed user types.
- Designed and deployed Learner and Trainer AI bots using FastAPI and GPT-4o to power career guidance and roadmap generation for end users.
- Implemented an intent classification pipeline (TF-IDF → DistilBERT) achieving 98.5% accuracy and a weighted F1 of 0.985 across 8 intents, with structured JSON extraction for downstream services.
- Architected a hybrid recommendation engine combining ML-based intent classification, LLM-driven JSON extraction, and weighted SQL filtering to deliver personalized, context-aware recommendations.
- Developed production-grade FastAPI services with stateful session management, rate limiting, validation, and error handling while resolving SonarQube issues and collaborating in a 10+ engineer Agile team.

HIDANI Tech

Remote

Lead Extraction Intern

Mar 2025 – Jun 2025

- Extracted and cleaned business lead data from multiple online sources, preparing structured datasets for downstream automation workflows.
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PROJECTS

Cafelytics – AI Analytics Platform | *Python, Django, Scikit-Learn, Random Forest, MySQL*

- Architected an AI-powered canteen analytics platform combining Random Forest models with Django to predict meal popularity and generate budget-aware recommendations.
- Integrated predictive analytics into a Django web application backed by MySQL, enabling data-driven canteen management and reduced food wastage.

AI Stress Detection System | *Python, TensorFlow, CNN, Pandas*

- Designed and trained a Convolutional Neural Network (CNN) in TensorFlow to classify stress levels from behavioral and health-related indicators.
- Engineered the data preprocessing and feature extraction pipeline, and evaluated model performance through visualization-driven analysis.

BizPilot AI | *Python, FastAPI, LLMs, WhatsApp Business API, MySQL*

- Designing conversational AI workflows on FastAPI and LLMs to automate customer support, appointment management, and business operations for small businesses.
- Conducting customer discovery research to validate product requirements and inform scalable system architecture for multi-industry deployment.

Student Performance Predictor | *Python, Django, Scikit-Learn*

- Developed a Machine Learning application using Scikit-Learn to predict student academic outcomes via classification models.
- Implemented data visualization and a real-time prediction interface through a Django web application.

TECHNICAL SKILLS

AI / ML: LLMs, RAG, Prompt Engineering, NLP, Machine Learning, Deep Learning, CNN, Scikit-Learn, TensorFlow, Recommendation Systems

AI Backend Engineering: FastAPI, REST APIs, Django, Python, SQL

Languages: Python, Java, SQL, C

Data Tools: Pandas, NumPy, Power BI, Data Visualization

Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, SonarQube

EDUCATION

Hyderabad Institute of Technology and Management (HITAM)

Hyderabad, Telangana

B.Tech, Artificial Intelligence & Machine Learning – CGPA: 9.13/10 (Branch Topper)

2023 – 2027

- **Relevant Coursework:** Machine Learning, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

CERTIFICATIONS

IBM Data Visualization with Python | Deloitte Data Analytics Job Simulation | NPTEL Programming with Java (Elite) | NPTEL Data Structures | Cisco PCAP – Programming Essentials in Python | TASK Python Certification | NxtWave Fellowship

ACHIEVEMENTS

- Branch Topper – B.Tech Artificial Intelligence & Machine Learning
- Selected for Smart India Hackathon (Internal Round)